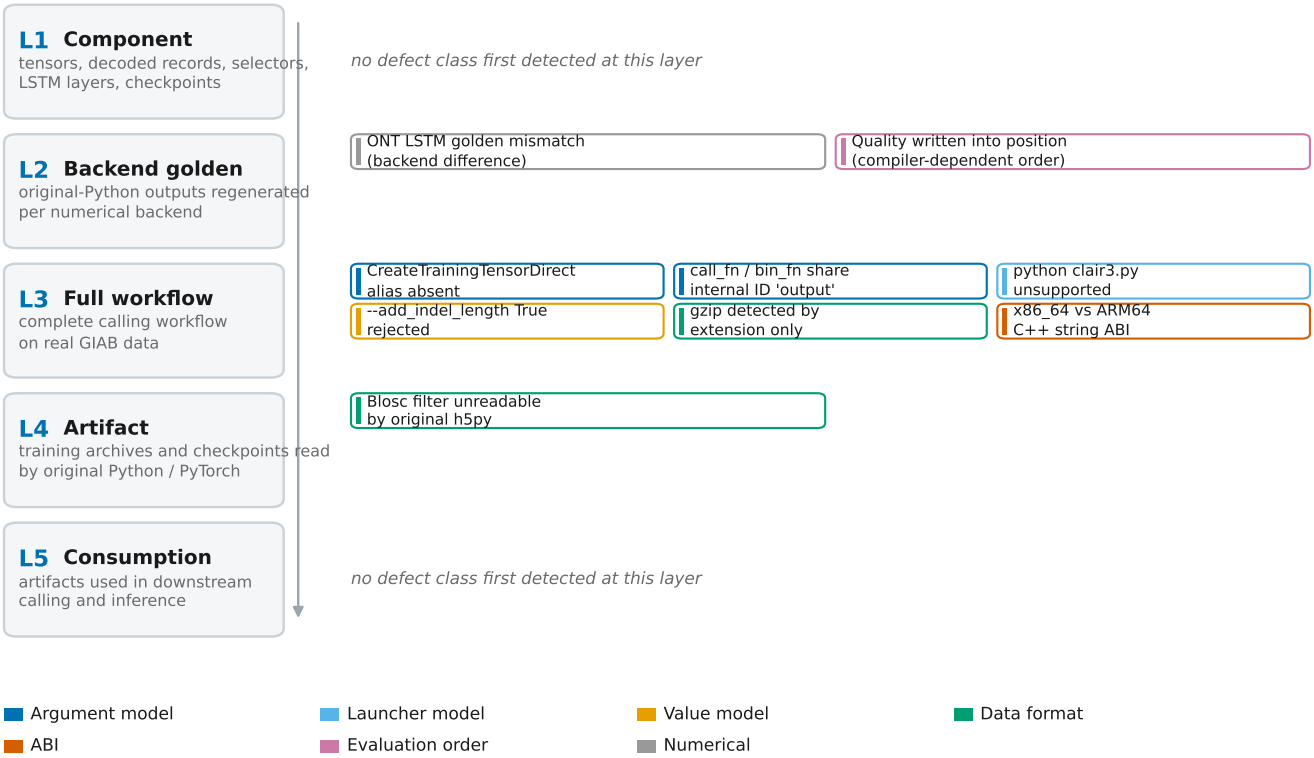
Figure 1. Differential-validation ladder for a scientific software rewrite

Nine compatibility defect classes, seven first detected only at L3 or beyond; none at L1.
Layers increase in scope from unit-level checks to scientific outputs.